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Generative AI: Changing How Business Innovates and Operates



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Generative AI Offers Us Much More Than Unique Visuals and the Best Words

This Person Does Not Exist

Text-to-Image: Stable Diffusion

[Stable Diffusion 2-1 - a Hugging Face Space by stabilityai](#)



Source: Wikimedia Commons

Generative AI Definition

Generative AI Generative AI refers to AI techniques that learn a representation of artifacts from data, and use it to generate brand-new, unique artifacts that resemble but don't repeat the original data.

Generative AI can produce totally novel content (including text, images, video, audio, structures), computer code, synthetic data, workflows and models of physical objects.

Generative AI also can be used in art, drug discovery or material design.



Generative AI Began to Quickly Emerge as a Topic on Earnings Calls in Early 2023

S&P Global 1200 Company CEOs

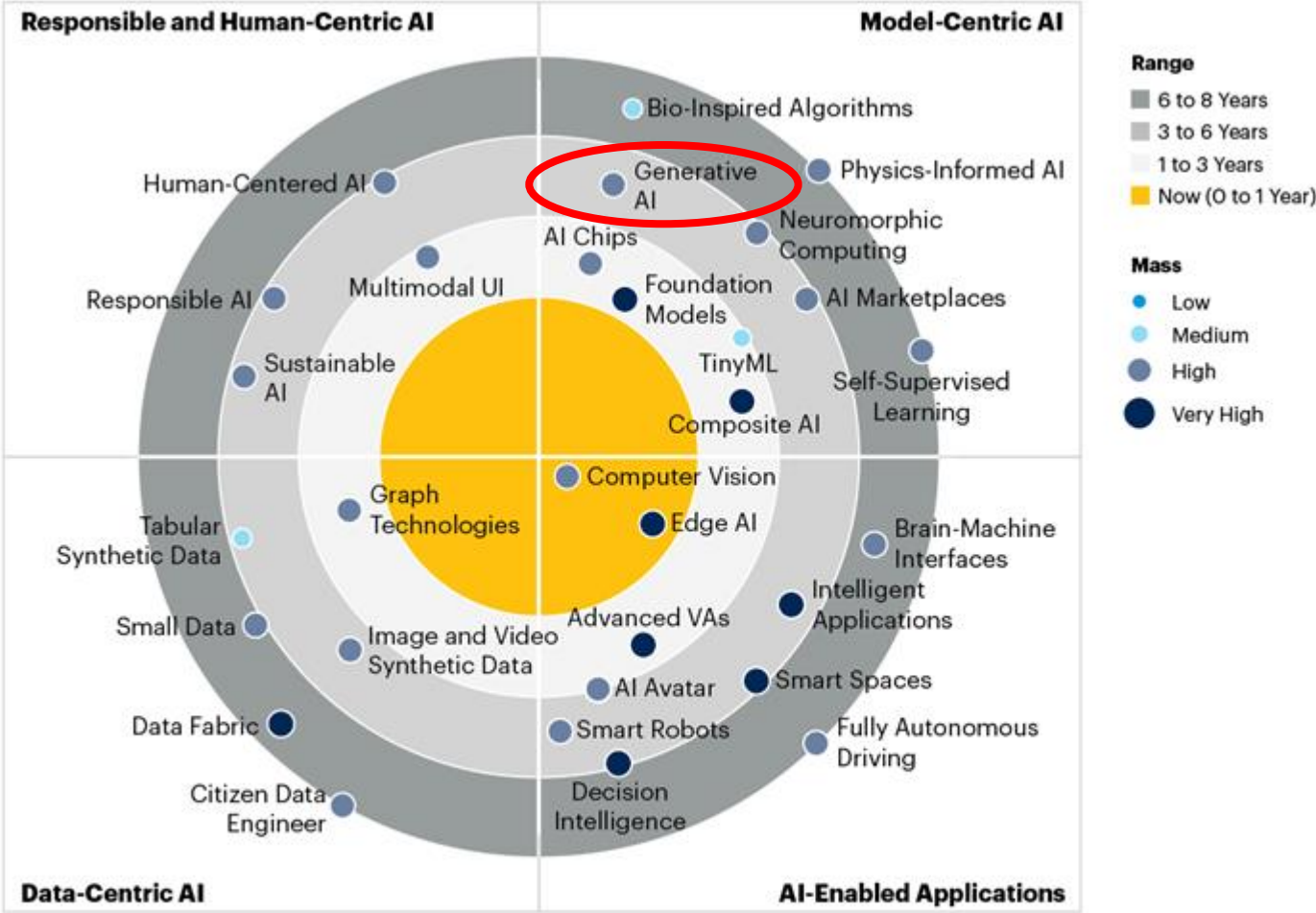
2% of CEOs discussed GenAI in their earning calls

1% of CEOs discussed deploying GenAI

(From January to April 2023)

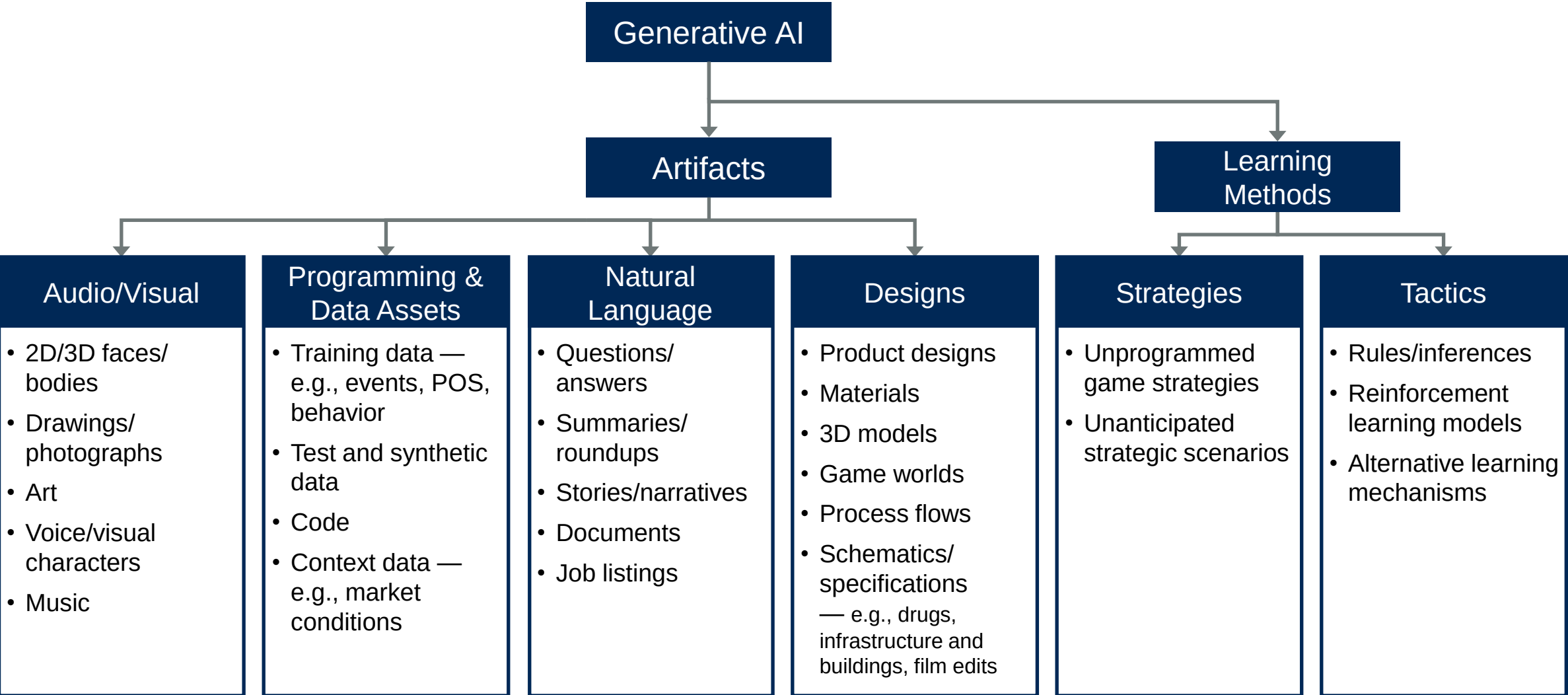
Company	CEO
Genpact	Nallicheri Vaidyanathan Tyagarajan President, CEO and Director
Etsy	Joshua G. Silverman CEO, President and Director
Accenture	Julie T. Spellman Sweet CEO and Chairman
Meta Platforms	Mark Elliot Zuckerberg Founder, Chairman and CEO
Cadence Design Systems	Anirudh Devgan CEO
Leidos Holdings	Roger A. Krone Chairman and CEO

Emerging Tech Impact Radar: Artificial Intelligence

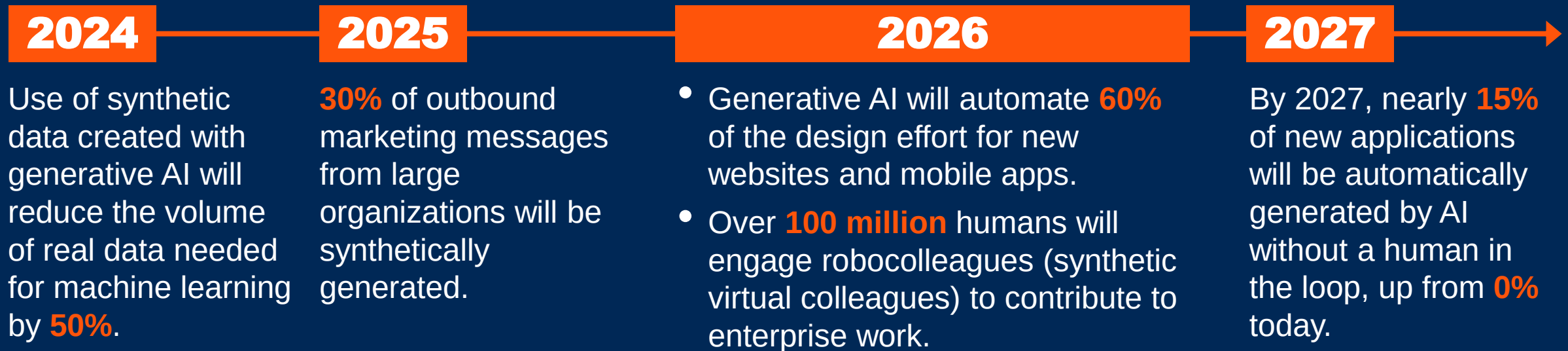


Source: Gartner
AI = artificial intelligence; TinyML = Tiny Machine Learning; UI = user interface; VA = virtual assistant
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Business Artifacts That GenAI Can Generate Today



Possible Scenarios and Trajectories for GenAI



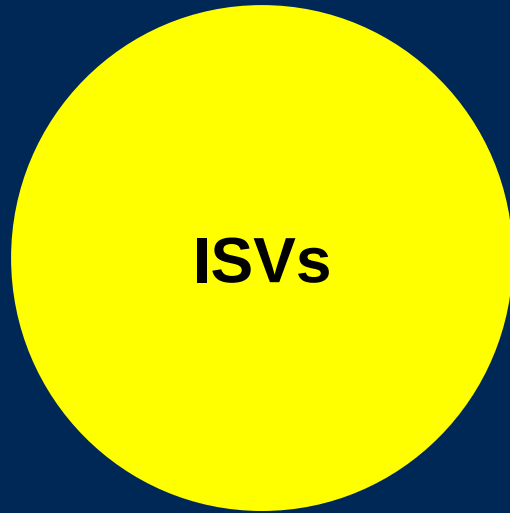
Source: [Predicts 2022: Generative AI Is Poised to Revolutionize Digital Product Development](#) 11 January 2022 - ID G00757883
By Brent Stewart, Van Baker, Joachim Herschmann

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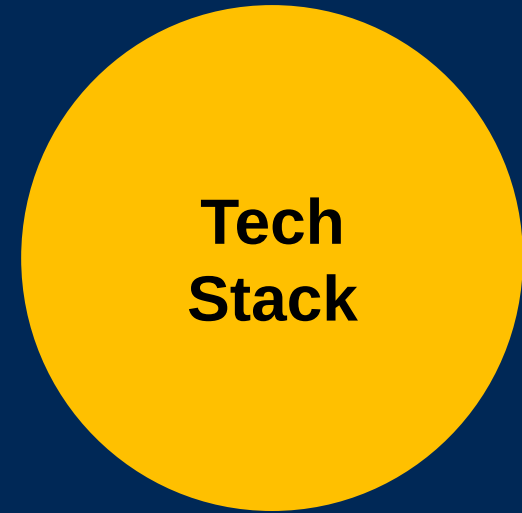
Tech & Service Provider Response to the Generative AI Opportunity



New applications, waves of new 'AI-Natives, Growing need for modernization



Business & IT Consulting, FMs-as-a-Service



FMs-as-a-Service, AIChips, vector database, HPC

ETT Case Based Research (CBR)

- A structured field study seeking to reveal ongoing technology and business demands for emerging technologies.
- Interviewed (30) vendors offering software/platforms enabled by generative AI.
- Collected nearly 100 case studies and related use cases.
- Field survey process spanned Q4-22 through Q1-23 (with a few additional engagement in Q2-23).
- Published research from the CBR study is currently available with several documents in process.

AI Avatars

AI Avatars Are Complex Technological Creations

AI Avatar Technology Layers and Components

AI Avatar Shell (CGI and 3D Real-Time Modeling)

Responsible AI

Emotion AI

Multimodal UI

Conversational AI

Speech Recognition

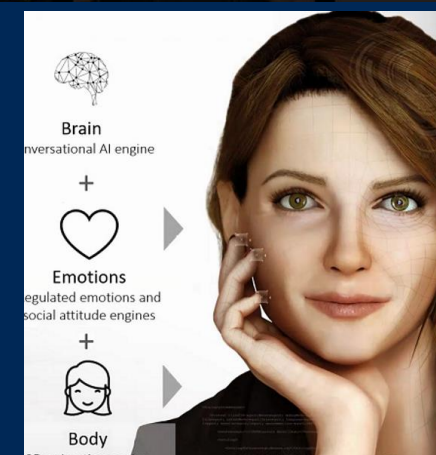
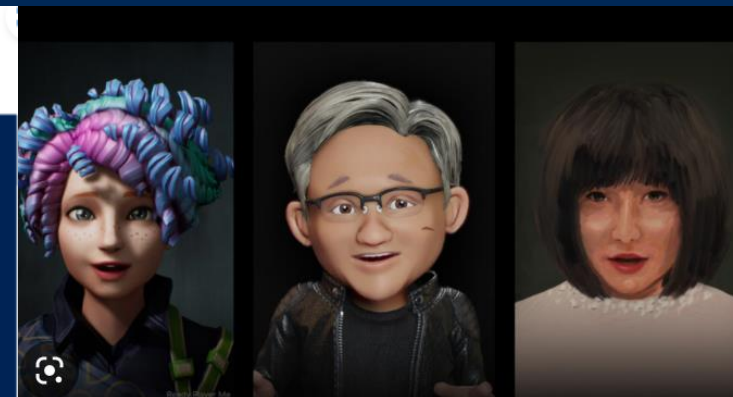
NLP
(NLU and NLG)

Machine Translation

Synthetic Voice

Note: NLP = natural language processing, NLU = natural language understanding, NLG = natural language generation

Source: Gartner



Sample Vendors: Amelia; DAVI; Openstream.ai (Eva); Kleon, NVIDIA (Omniverse Avatar Cloud Engine); Rephrase.ai, Soul Machines; Synthesia, Uneeq

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Advanced in Generative AI Speed Up Creation of AI Avatars and Open Up New Use Cases



Use Case examples:

- Customer support representatives
- Virtual salesperson or financial advisor
- Supporting hybrid or virtual coworking and collaboration.
- Customer assistant in a kiosk in an airport or a social venue.
- Training assistant (possibly in a metaverse) for employees or students.
- Virtual presence at entertainment events, like a virtual concert or gallery.

By 2025, AI avatars leveraging text-to-video using generative AI technology will support 70% of digital and marketing communications from less than 5% in 2022.

AI Avatar Implications & Recommendations

- Implications

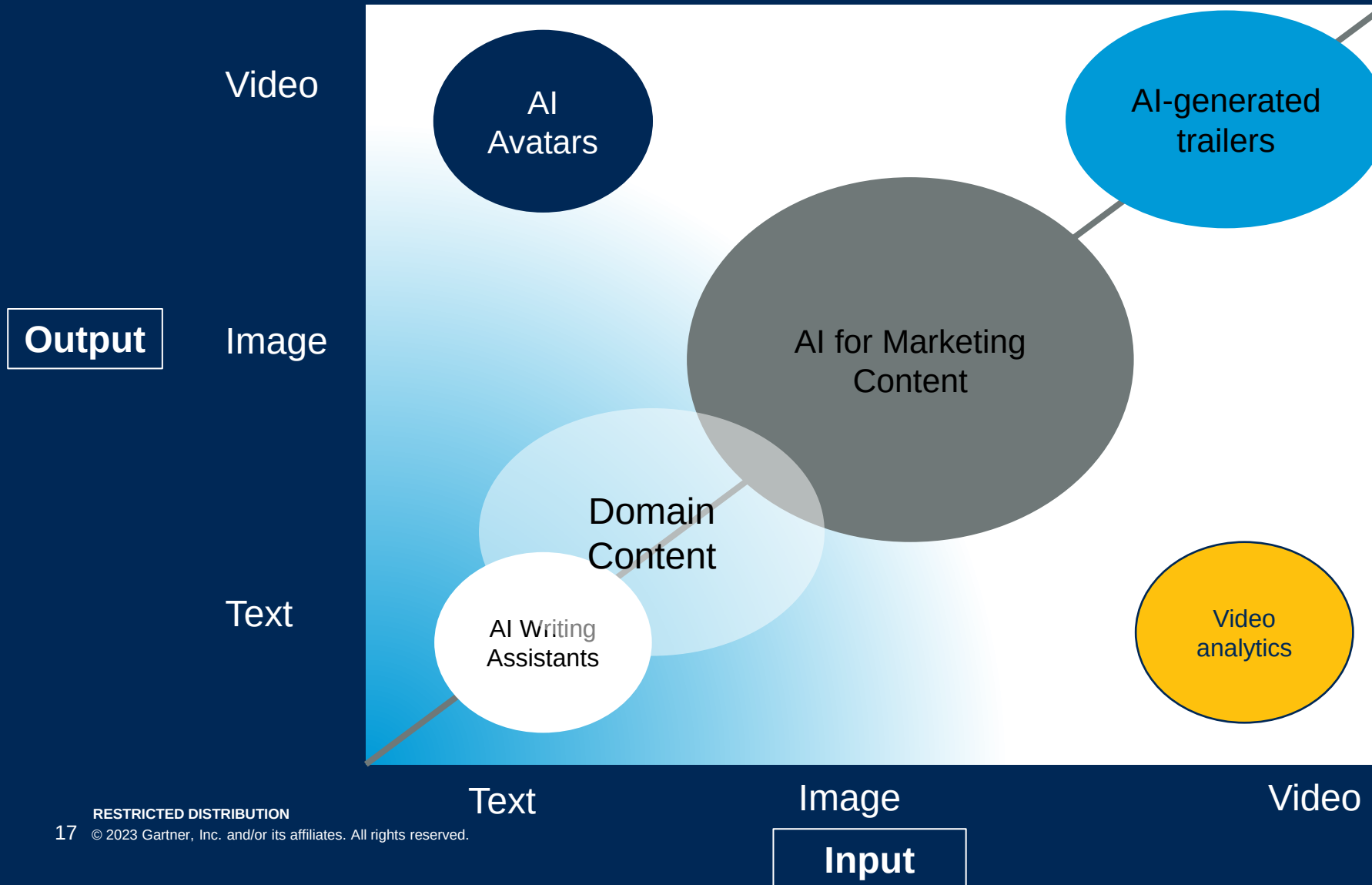
- AI avatars leveraging **text-to-video** will enhance the creativity and humanize internal and customer communications, enabling hyper personalization.
- **Two biggest technical challenges** are combining various emerging technologies to enable realistic, human-centered and super responsive AI avatars and applying this at scale.
- Concerns about deepfakes, nascent but lagging legal protection for users and content, and responsible application of AI principles are **challenges to scaling** future AI avatar deployments.

- Recommendations

- Identify use cases where AI avatars can add quantifiable business value to your software products, as brand ambassadors, digital customer care reps or even in the metaverse.
- Further enhance VA offerings by integrating synthetic voice and emotion AI technology into voice interfaces, like AI avatars in kiosks, virtual training or immersive environments.
- Create visually realistic AI avatars by determining which technological elements can be developed internally, such as avatar CGI shells, and by collaborating with pertinent NLT providers for features like synthetic voice, multimodal AI or emotion AI.

AI-Generated Content

AI-Generated Content – Market Landscape



Use Case For Text Automation

- General – Q&A, Translation, Summarization
- Domain specific
 - Summarization and Translation of Legal Document
 - Automated Report Writing from Clinical Trial Data in Pharma
 - Business Model, Market Sizing, SWOT Analysis, PEST Analysis, Brand Naming
 - Media by Converting Sports Data Into News
 - Sports Player Performance Report From Video Analytics
 - Real Time Market Insights From Financial Market Data

Use Case: Automated Clinical Trial Reports

Business Challenge

- To reduce third party dependency to build reports from clinical trial data.
- To reduce time, cost and effort required to build one report.
- To reduce human error and operational efficiency.

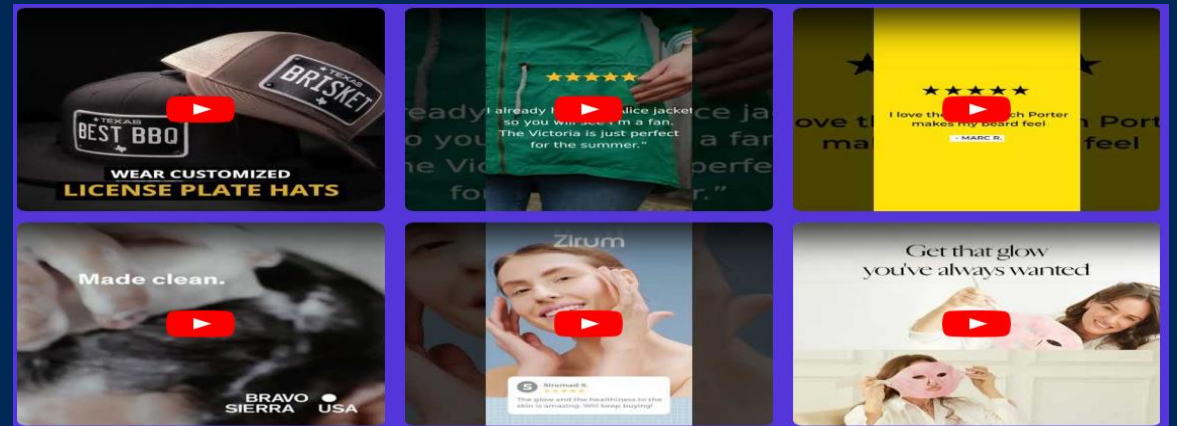
Business Value Achieved

- 65% reduction in time spent on documentation generation hence faster approvals.
- 40% expected cost reduction on medical writing.
- 50% reduction in reviews required.
- Increase operational efficiencies with more standardized and streamlined process.

Source <https://www.narrativa.com/4-benefits-of-automating-clinical-study-reports-csrs/>, <https://www.narrativa.com/narrativa-lls-bring-ai-generated-reporting-to-the-clinical-trial-process/>

Use Cases For Image and Video Automation

- Personalized and Customized Advertisement Content based on Geography, Race, Age Group, Profession
- Creative Advertisement for Marketing Across Different Channels
- AI-generated customized video trailer based on preference towards emotions, thriller, action etc.



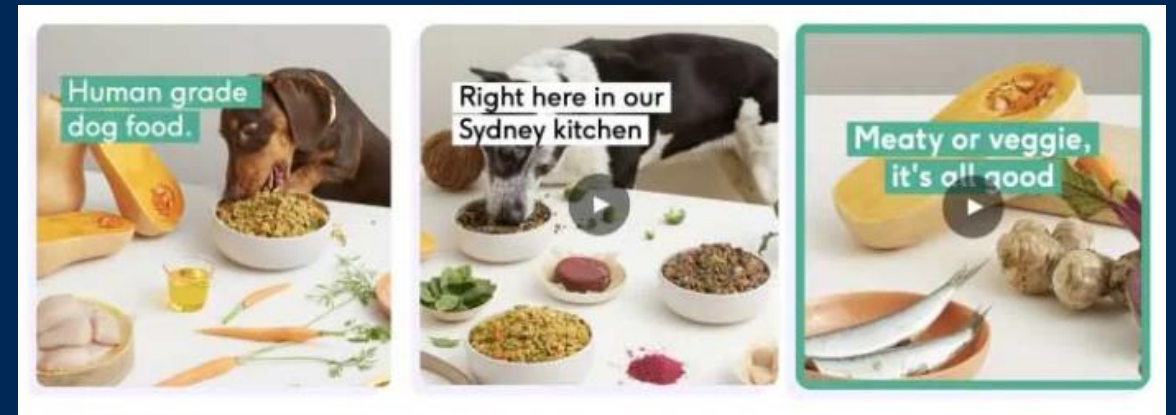
Use Case: Creative Ads Using Generative AI

Business Challenge

- To reduce cost per acquisition by 50% and find the best performing creatives ads for a dog food brand.

Business Value Achieved

- 16 ads tested to find the best performing ads.
- Average 15% reduction in cost per acquisition.
- Max 56% reduction in cost per acquisition.



Source <https://www.trypencil.com/mini-case-studies>

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AI-Generated Content Implications & Recommendations

- Implications
 - By 2027, Nearly 50% of outbound marketing messages from large organizations will be synthetically generated; up from less than 5% today.
 - Broad spectrum marketing and advertising support: copy, images, video driving massive disruption to internal and third-party marketing ecosystems.
 - Same technology can be put to misuse and can lead to creation of fake news and propaganda.

AI-Generated Content Implications & Recommendations

- Recommendations
 - Build transformational generative AI domain-specific solutions by applying generative AI to information-driven processes.
 - Include veracity and validation as part of your offerings in line with the risk profile of your key business use cases.
 - Ensure responsible and ethical use of AI by being transparent. Amend privacy disclosures to include information about the role these tools play in shaping digital experience.

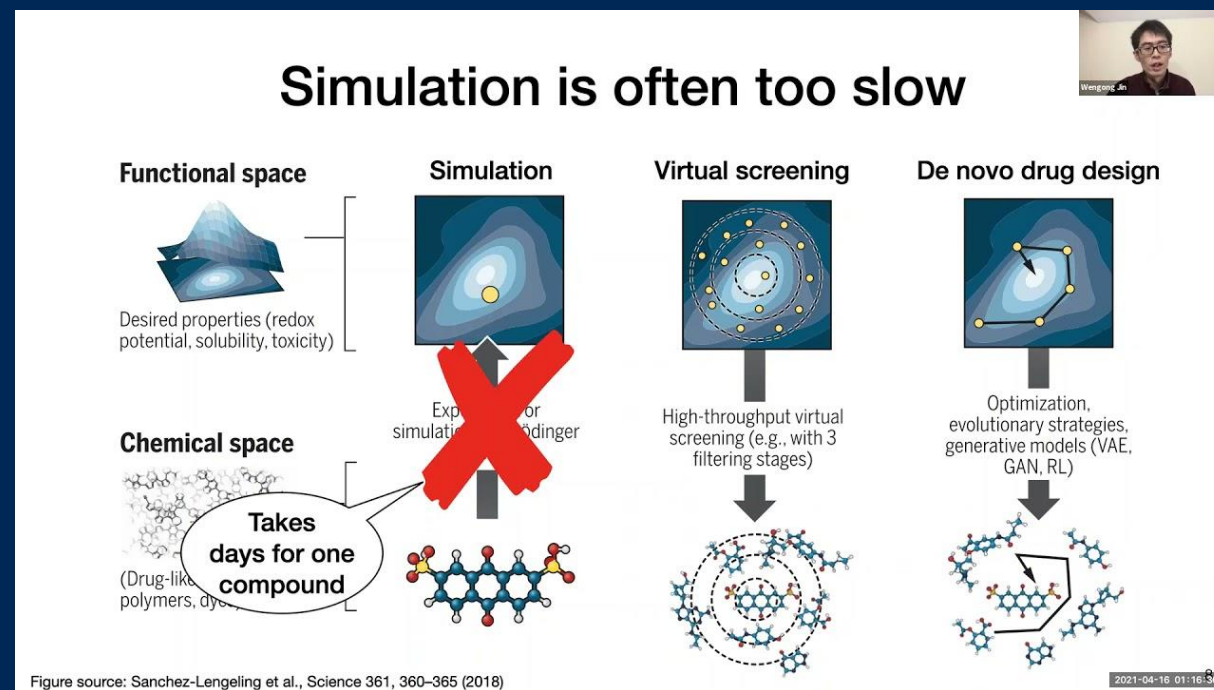
Novel Molecule & Compound Discovery

Generative AI for Novel Molecules and Compounds

Deep generative models, such as:

- Generative adversarial networks (GANs)
- Variational autoencoders (VAEs), and,
- Deep reinforcement learning algorithms,

are utilized to generate novel molecules, optimize existing compounds, and predict properties and activities of potential drugs, chemicals and materials.

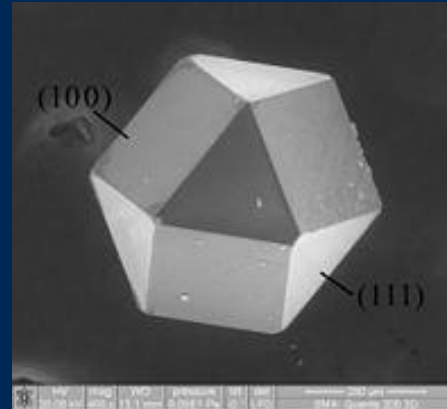


Novel Molecule and Compounds Use Cases

Drug discovery



Material science



Representative vendors

- Standigm
- Iktos
- Chempass
- IBM
- Insilico



Food & Beverage



Consumer Packed Goods

Novel Molecule and Compounds Implications & Recommendations

- **Implications**

- Faster, cheaper and expanded discovery potential to massively advance human health; Lower costs mean pharma and biotech make more bets
- Massive fundraising and less cost-intensive in vitro work are lowering capital barriers for startup discovery efforts.
- A fast-growing body of treatments entering preclinical trials over the last five years.

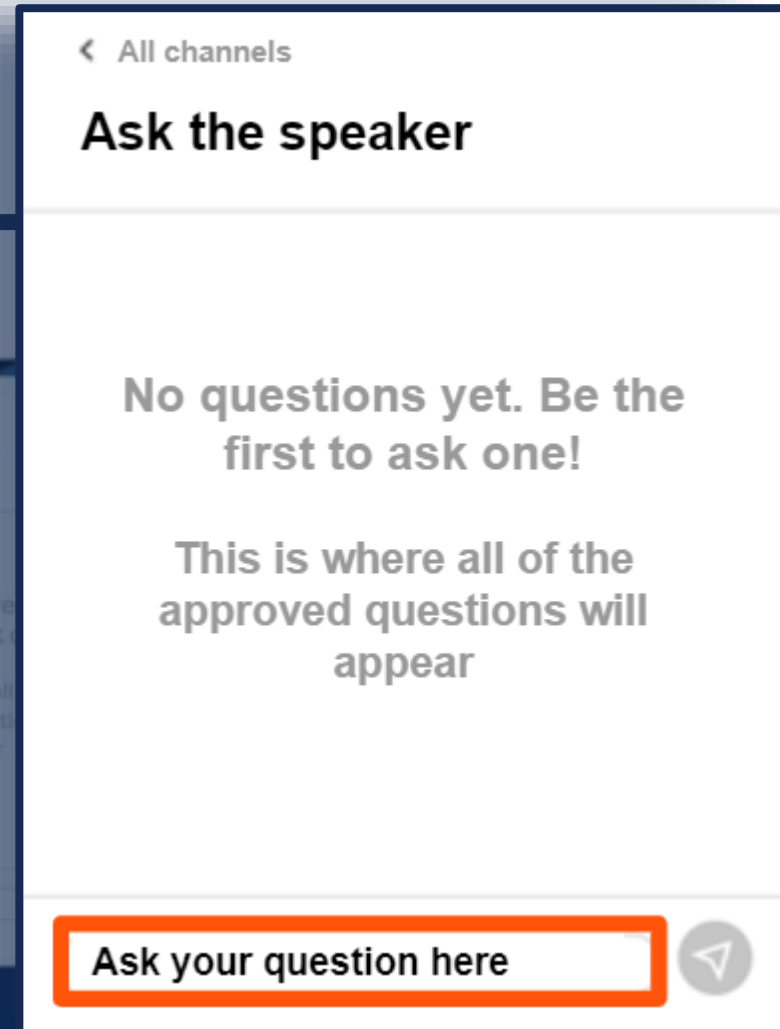
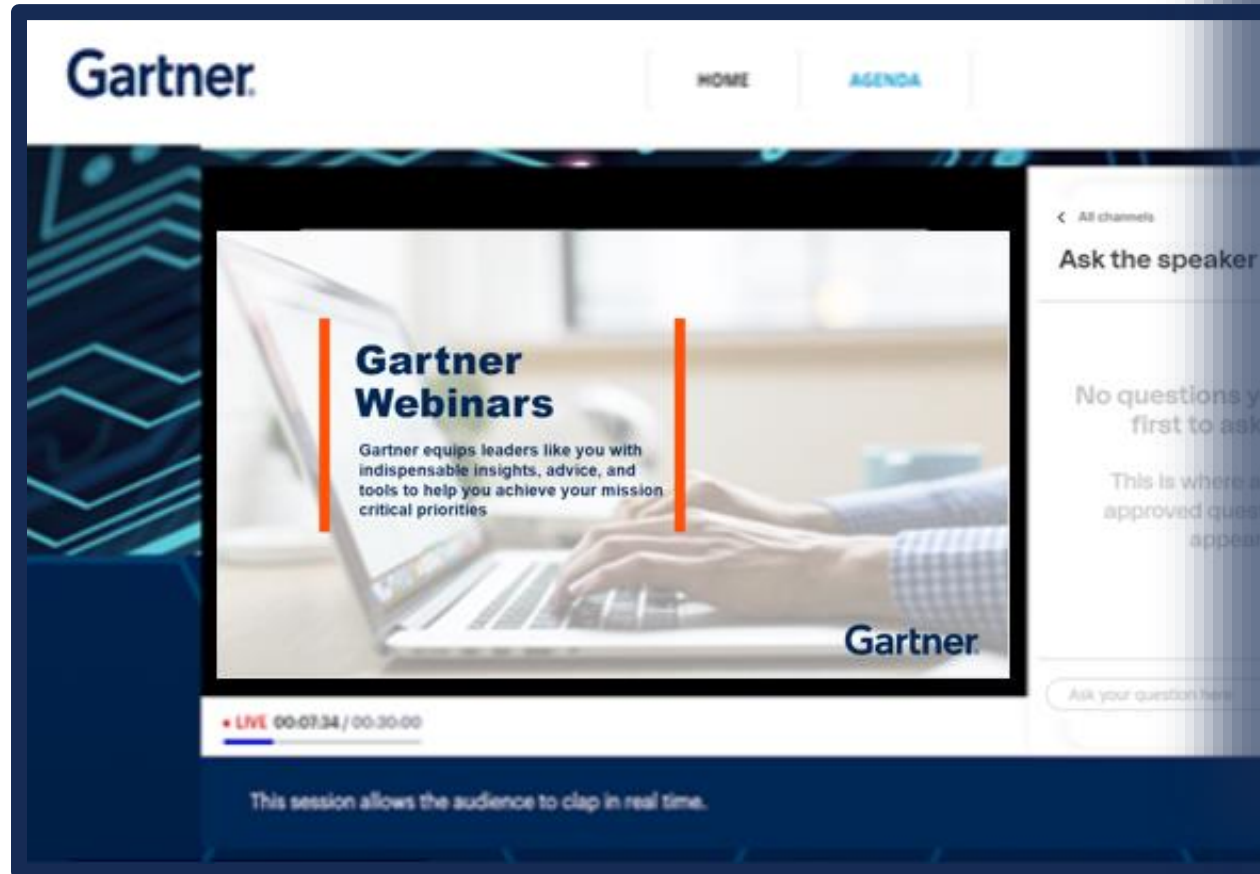
- **Recommendations**

- Selling picks and shovels for the gold rush, is the new gold rush. Technology stacks, consulting and managed services for Foundation Models (as-a-Service).
- Capitalize on the value of IP generated and secure contingent revenue from royalties, out-licensing, JVs or becoming asset-owning.
- Pursue specific, high-value use cases that are directly aligned with specific discovery programs.

Recommendations

- Focus on potential use cases that are most important to your installed base, and prioritize specialized industry and domain solutions when incorporating generative AI in your products.
 - The key to success is to quantify value and impact for your prospects.
- Develop new product and service catalogs focused on existing customer data assets to operationalize Foundation Model-based value.
- Aggressively partner with innovative, generative- native companies to accelerate customer value.
- Investigate an acquire strategy which offers intellectual property and scarce skillsets.

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Top Trends for Tech Providers in 2023

Explore the top trends product leaders must evaluate across all dimensions of their business. These trends are the effects of three broad themes grouped below:

- Increased reliance on tech
- Opportunities through new tech
- Impact of macro forces

→ **Download the Research**

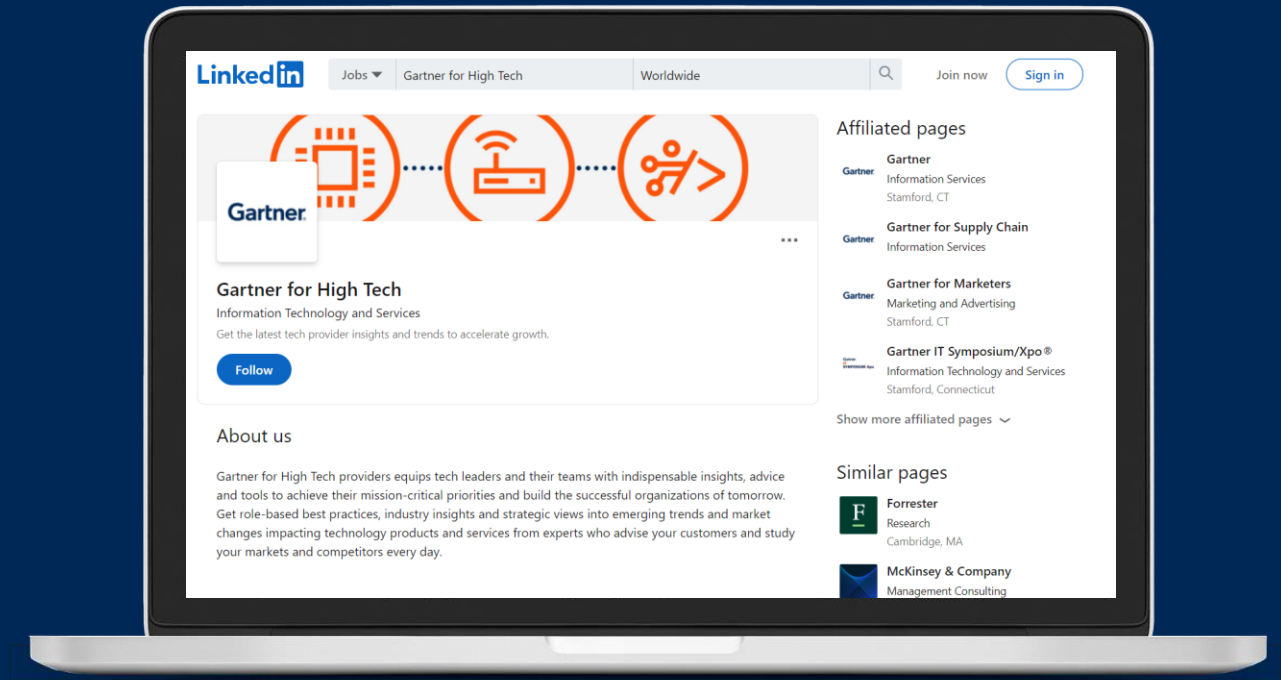


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As cloud computing evolves from technology enabler to business disruptor, IT leaders must ensure they understand their organization's business strategy. Only then can they seek opportunities to leverage new and emerging cloud capabilities to accelerate that strategy. This free webinar reveals Gartner's top predictions for where cloud computing will be by 2027, and explores how these predictions will shape your cloud value proposition.

- Explore what cloud computing will look like in 2027
- Discover how multi-cloud and cloud native can affect organizations' cloud efforts
- Ensure a successful cloud journey for your organization

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Speakers